Category-Aware Explainable Machine Learning for Urban Service Demand Forecasting in Smart Cities

Tran Dai Phong Lam[1*], Thu Le[1], Nguyen Quoc Hung[1], Nguyen Trung Trinh[1]

[1*]FPT University, Ho Chi Minh City, Vietnam.

*Corresponding author(s). E-mail(s): phonglam2599@gmail.com;
Contributing authors: thulvm@fpt.edu.vn; hungtvt2222@gmail.com; trinhnguyen112355@gmail.com;

Abstract

Urban 311 systems record large-scale traces of reported municipal demand, but forecasting operationally useful surges remains difficult because reports are spatially uneven, service categories differ, and reporting behavior shapes observed counts. We formulate next-week abnormal reported-demand forecasting as a neighborhood-week-service-category early-warning task. Using New York City as a case study, the analysis integrates 30.9 million NTA-matched 311 requests with shifted historical demand features, calendar timing, NOAA weather exposure, OpenStreetMap point-of-interest context, and PLUTO built-environment descriptors. A target-week chronological protocol trains on 2015-2022 target weeks, validates on 2023, and tests on 2024-2025. The validation-selected decision strategy averages LightGBM and XGBoost scores and applies complaint-category-specific thresholds selected only on validation. On the held-out future test period, it obtains F1 = 0.3839, precision = 0.3071, recall = 0.5117, PR-AUC = 0.3365, and ROC-AUC = 0.7720. SHAP analysis shows that historical temporal demand features dominate, while service category, built environment, POI context, weather, and calendar features add interpretable signal. The system is positioned as decision support for prioritizing elevated reported-demand risk, not as a causal model of all urban conditions.

Keywords: urban computing, smart cities, 311 service requests, service-demand forecasting, explainable machine learning, SHAP, spatiotemporal modeling

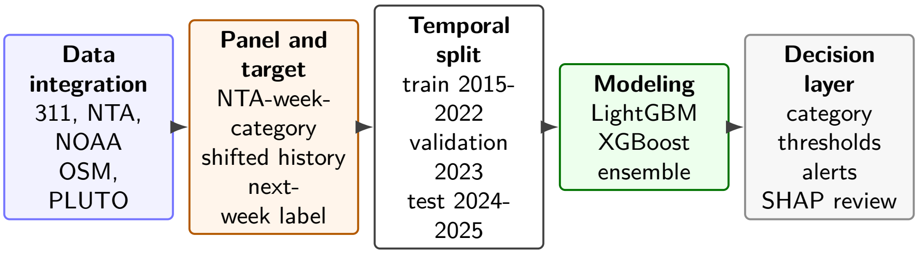

Fig. 1 Overview of the leakage-controlled temporal pipeline. The modeling target is next-week abnormal reported demand for each NTA, week, and service category; OSM and PLUTO are used as static context snapshots.

1 Introduction

Urban service systems increasingly rely on administrative and citizen-generated data to monitor conditions, allocate attention, and anticipate operational pressure. NYC 311 is especially valuable because it records requests across infrastructure, housing, sanitation, public safety, environmental, and quality-of-life domains. Yet 311 counts are not direct measurements of all underlying urban problems. They reflect both conditions and the propensity to report them, which varies with access, awareness, trust, service expectations, and governance processes [1–4]. Forecasts built from these data therefore need careful wording: the target is abnormal reported service demand, not the true incidence of every urban issue.

We study a prospective early-warning task at the neighborhood tabulation area (NTA), week, and complaint-category level. For each feature week, the model estimates whether the following week will show abnormal reported demand relative to recent local history. The design combines historical 311 demand, calendar timing, weekly weather exposure, OpenStreetMap (OSM) point-of-interest context, and PLUTO built-environment descriptors. The target and temporal features are leakage-controlled: lags and rolling statistics are shifted so only information available before the prediction week is used. Static OSM and PLUTO snapshots are treated as contextual descriptors and are discussed as a snapshot/look-ahead limitation rather than as dynamic covariates.

The analysis examines whether next-week abnormal reported demand can be forecast under a strictly chronological future test, how service-specific thresholds alter precision, recall, and alert volume across complaint domains, and which temporal and urban-context features contribute most strongly to model predictions. Its contributions are a leakage-controlled NTA-week-category forecasting task, a large multi-source NYC panel built from 30,940,129 NTA-matched 311 requests and public contextual data, category-specific alert calibration, prospective target-week evaluation, and explainability that separates predictive association from causal interpretation. The practical aim is to prioritize analyst attention toward elevated reported-demand risk without treating 311 data as a complete map of urban need. Figure 1 summarizes the workflow.

2 Related Work

Urban computing links administrative, geographic, and sensed data to support city analysis and decision making [5]. Recent spatiotemporal learning surveys emphasize the value of temporal structure, spatial context, and prospective validation in urban prediction problems [6]. Within 311 research, prior work shows that complaint structure can distinguish urban locations [1], while reporting bias can distort data-driven governance decisions [2, 3]. Related domain studies connect 311 requests to flooding [7], extreme heat [8], foodborne illness signals [9], open-data curation [4], and municipal service-level design [10]. Our task differs by forecasting future abnormal reported demand across multiple service categories in a single citywide panel.

Contextual urban features can improve service-demand models when interpreted carefully. OSM provides volunteered geographic information and POI records useful for describing neighborhood function, but coverage and tagging are uneven [11, 12]. POI and land-use features have been used for urban change, functional-zone, land-use, and traffic-flow modeling [13–16]. In this work, OSM POIs and PLUTO built-environment summaries support predictive context; they are not interpreted as contemporaneous causal drivers.

The forecasting target is an imbalanced binary event, making ordinary accuracy uninformative. Imbalance-learning surveys and evaluation work motivate precision, recall, F1, PR-AUC, ROC-AUC, and balanced accuracy as complementary views [17–21]. Gradient-boosted tree methods such as XGBoost and LightGBM are strong tabular baselines for nonlinear interactions [22, 23]. Because operational use requires transparency, we use SHAP and TreeSHAP to explain a tuned LightGBM component while following cautions that explanations of black-box models should not be mistaken for causal guarantees [24–26].

3 Methodology

3.1 Data and Task Formulation

The project integrates six public data sources: NYC Open Data 311 Service Requests from the historical 2010–2019 archive and the current 2020–present archive [27, 28], 2020 NTA boundaries [29], NOAA GHCN-Daily weather for Central Park station `GHCND:USW00094728` [30], OSM POIs queried through Overpass [31], and NYC PLUTO/MapPLUTO land-use records [32]. The analysis window uses 2015–2025 records from the combined 311 archives. Weekly 311 counts use Monday week starts and are spatially joined to 2020 NTAs. Weather covariates summarize the feature week t, not the target week $t + 1$. OSM and PLUTO variables are fixed extraction snapshots from 2026-06-25 and are used as static context descriptors.

The analytical unit is (n, t, c), where n is an NTA, t is a feature-week start date, and c is one of nine stable complaint categories: noise, housing, water/sewer, public safety, parking/traffic, environment, infrastructure, sanitation, and other. These categories reduce sparsity while preserving service meaning. The dense panel contains 262 NTAs, 575 weekly periods, and nine categories, yielding 1,355,850 rows. Relative

Table 1 Dataset construction summary.

Item	Value	Item	Value
NTA-matched 311	30,940,129	Match rate	99.9777%
Unit	NTA-week-category	Panel rows	1,355,850
Valid-label rows	1,334,628	Model-ready rows	1,325,196
Excluded rows	30,654	NTAs	262
Weeks	575	Categories	9
Coverage	2014-12-29-2025-12-29	NOAA days	4,018
OSM POIs	77,083	Output columns	252
Predefined schema	215 inputs	Final feature subset	213 inputs

to the full dense panel, 30,654 rows are excluded due to warm-up/history requirements and unavailable next-week targets, leaving 1,325,196 model-ready rows. Table 1 summarizes the construction.

For each feature week t, all predictors are available after that week closes. The target week is $t+1$, seven days after the feature-week start date:

$$y_{n,t,c} = \mathbf{1}\{\text{count}_{n,t+1,c} > \mu^{8w}_{n,t,c} + 1.5\sigma^{8w}_{n,t,c}\}.$$

The rolling mean $\mu^{8w}_{n,t,c}$ and standard deviation $\sigma^{8w}_{n,t,c}$ are computed from a shifted historical eight-week window. The label therefore identifies next-week abnormal reported demand relative to each local recent baseline.

3.2 Predictive Models and Category-Aware Decision Layer

The model-ready configuration starts from a predefined 215-input schema specified before model fitting. The final tuned comparison uses a 213-input no-COVID-period subset, excluding two period indicators so performance does not rely on COVID-era labels; one-hot encoding expands this to 229 model columns. Historical features include lag counts, rolling statistics, ratios to historical baselines, current demand, and history-availability flags. Calendar, weather, OSM, and PLUTO features add temporal and neighborhood context.

We compare rule baselines, logistic regression, LightGBM, XGBoost, and an equal-weight LightGBM/XGBoost ensemble. Numeric medians and categorical levels are fitted on the training period only. Hyperparameters are selected from stratified 300,000-row tuning runs for reproducibility, then final scores are rebuilt on all 954,990 training rows. Table 2 records the compact protocol.

The final settings are read from the LightGBM, XGBoost, and ensemble run summaries under `data/processed/model_results/`. Parameters not explicitly passed by the scripts, including XGBoost `gamma` and `reg_alpha`, retain their library defaults.

Let $s^{L}_{n,t,c}$ be the tuned LightGBM score and $s^{X}_{n,t,c}$ the tuned XGBoost score. The ensemble score is $s^{E}_{n,t,c} = 0.5s^{L}_{n,t,c} + 0.5s^{X}_{n,t,c}$. The category-aware decision layer assigns each category c a validation-selected threshold τ_c, producing $\hat{y}_{n,t,c} = \mathbf{1}\{s^{E}_{n,t,c} \geq \tau_c\}$. This is category-aware at the decision stage: the same probability score can map to different alert decisions because service domains differ in score distribution, prevalence, and operating cost.

Table 2 Compact tuning and final-score protocol for gradient-boosted models.

Item	Setting
Feature set	213-input no-COVID-period subset; one-hot expansion to 229 model columns
Tuning sample/seed	Stratified train sample, `max_train_rows` = 300,000, seed 42; selected candidates rebuilt on all 954,990 training rows
LightGBM grid	Depth/leaf/lr: (8,63,0.05), (6,31,0.03), (10,127,0.03); class weights none, square-root imbalance, half imbalance, and full balanced weight
XGBoost grid	Depth/lr: (8,0.05), (6,0.03), (5,0.03); class weights none, square-root imbalance, half imbalance, and full balanced weight
Selection	Validation F1; selected LightGBM and XGBoost both use 800 estimators, lr 0.03, max depth 6, and square-root positive weight 2.6216; no early stopping is used
Selected settings	LightGBM: binary objective, num_leaves = 31, min_child_samples = 100, subsample = 0.85, colsample_bytree = 0.85, reg_lambda = 3.0, class_weight = $\{0 : 1, 1 : 2.6216\}$. XGBoost: binary:logistic objective, eval_metric = aucpr, min_child_weight = 50, subsample = 0.85, colsample_bytree = 0.85, reg_lambda = 3.0, scale_pos_weight = 2.6216, tree_method = hist. Seed 42 is used for sampling and model fitting
Thresholds	Validation grid 0.01–0.99 step 0.005 plus score quantiles; category fallback to global threshold if validation positives < 30

3.3 Evaluation and Explainability Protocol

Evaluation uses a chronological split by target week: 2015-2022 for training, 2023 for validation, and 2024-2025 for testing. The model-ready split contains 954,990 train rows with 121,306 positives, 122,616 validation rows with 15,574 positives, and 247,590 test rows with 31,570 positives. Validation and test positive shares are approximately 12.7%. All thresholds are selected on validation only and evaluated unchanged on the held-out test period.

F1 is the primary validation-ranking metric because the alerting task requires balancing event detection and alert conversion. We also report precision, recall, PR-AUC, ROC-AUC, balanced accuracy, alert rate, and confusion matrices. Threshold tuning is treated as a method component because it changes hard-alert metrics and alert volume; it does not change PR-AUC or ROC-AUC for a fixed score vector.

For explainability, SHAP is computed for the tuned LightGBM component. Since the final evaluated score is an ensemble with category-specific thresholds, SHAP values are interpreted as evidence about a representative tree component under the same feature design and chronological protocol, not as a direct explanation of every final alert or as causal attribution.

4 Results

4.1 Overall Performance and Ablation Results

Rule and logistic baselines confirm that simple history alone is not enough for the final alerting task. On the 2024-2025 target-week test period, current-week persistence obtains F1 = 0.3111 and PR-AUC = 0.2359, a seasonal lag-52 margin obtains F1

Table 3 Feature-group ablation under the same target-week chronological protocol. Thresholds are selected on validation; PR-AUC and ROC-AUC use raw scores.

Feature configuration	Val PR-AUC	Test F1	Prec.	Rec.	Test PR-AUC	Test ROC-AUC
Core history + identifiers	0.2702	0.3513	0.2595	0.5434	0.2838	0.7374
Core + calendar	0.3072	0.3785	0.2873	0.5546	0.3262	0.7635
Core + calendar + weather	0.3143	0.3767	0.2843	0.5581	0.3246	0.7615
Core + calendar + weather + OSM	0.3189	0.3818	0.2915	0.5535	0.3314	0.7673
Core + calendar + weather + PLUTO	0.3165	0.3813	0.3008	0.5208	0.3299	0.7668
Full context	0.3180	0.3819	0.3066	0.5061	0.3297	0.7679

Table 4 Final decision-layer comparison. Thresholds are selected on validation and evaluated unchanged on the future test period.

Model	Threshold	Val F1	Test F1	Prec.	Rec.	PR-AUC	ROC-AUC	Alert rate
Ensemble LGBM(0.50)+XGB(0.50)	Per category	0.3809	0.3839	0.3071	0.5117	0.3365	0.7720	0.2124
Tuned XGBoost	Per category	0.3806	0.3845	0.3058	0.5175	0.3367	0.7722	0.2158
Tuned LightGBM	Per category	0.3802	0.3836	0.3022	0.5249	0.3346	0.7709	0.2214
Ensemble LGBM(0.50)+XGB(0.50)	Global	0.3777	0.3865	0.3001	0.5426	0.3365	0.7720	0.2305

= 0.2423 and PR-AUC = 0.1773, and the strongest rule baseline, a rolling z-score rule, obtains F1 = 0.3332 and PR-AUC = 0.2658. Logistic regression with historical features obtains F1 = 0.3256 and PR-AUC = 0.2598.

Table 3 reports a controlled LightGBM ablation using the same target-week split, validation-selected threshold policy, and 300,000-row stratified training sample for each configuration. The core history-and-identifier configuration already provides substantial predictive information, with test PR-AUC = 0.2838. Adding calendar terms raises test PR-AUC to 0.3262, while weather alone does not improve that run. OSM and PLUTO descriptors provide complementary context: the OSM configuration has the highest test PR-AUC among ablations (0.3314), and the PLUTO configuration yields higher precision with lower recall. The full context model has the strongest ROC-AUC and similar F1 to the OSM/PLUTO variants, so contextual features are best interpreted as incremental predictive signal rather than causal explanations.

Feature-set composition is fixed by the feature-set JSON. The core configuration retains complaint category, borough, current `complaint_count`, lag and rolling features, and history-availability indicators. Calendar and weather are added cumulatively; OSM and PLUTO are evaluated separately after weather, while the full-context setting includes both.

Table 4 compares final decision strategies. Following the validation-selection rule, the selected category-specific strategy is the LightGBM/XGBoost ensemble with complaint-category thresholds. On the held-out future test period, it achieves F1 = 0.3839, precision = 0.3071, recall = 0.5117, PR-AUC = 0.3365, ROC-AUC = 0.7720, and balanced accuracy = 0.6715. The global-threshold ensemble has slightly higher test F1, but it applies one alert rule across service domains; the category-specific layer is retained because it exposes service-specific operating points selected only on validation.

The final test confusion matrix contains 16,154 true positives, 36,440 false positives, 15,416 false negatives, and 179,580 true negatives. The model captures about half of true next-week abnormal reported-demand events while roughly three out of ten alerts correspond to true events. The top 1% of scored test rows has precision 0.5965 and 4.68-times lift over the test base rate; the top 5% has precision 0.4548;

Table 5 Final model test performance by complaint category on the 2024-2025 target-week period.

Category	Thr.	Positives	Alerts	PR-AUC	F1	Precision	Recall
Noise	0.4150	3,982	5,729	0.4229	0.4578	0.3880	0.5583
Public safety	0.3232	3,536	6,482	0.3312	0.4097	0.3166	0.5803
Housing	0.3850	3,720	5,412	0.3860	0.4025	0.3396	0.4941
Water/sewer	0.3700	3,462	6,090	0.3642	0.3938	0.3089	0.5433
Other	0.3314	3,620	6,695	0.3237	0.3762	0.2898	0.5359
Parking/traffic	0.3596	3,586	5,811	0.3255	0.3708	0.2998	0.4858
Environment	0.3385	2,896	5,924	0.3105	0.3633	0.2704	0.5532
Infrastructure	0.3814	3,596	5,900	0.2764	0.3484	0.2803	0.4600
Sanitation	0.3500	3,172	4,551	0.2570	0.3165	0.2685	0.3852

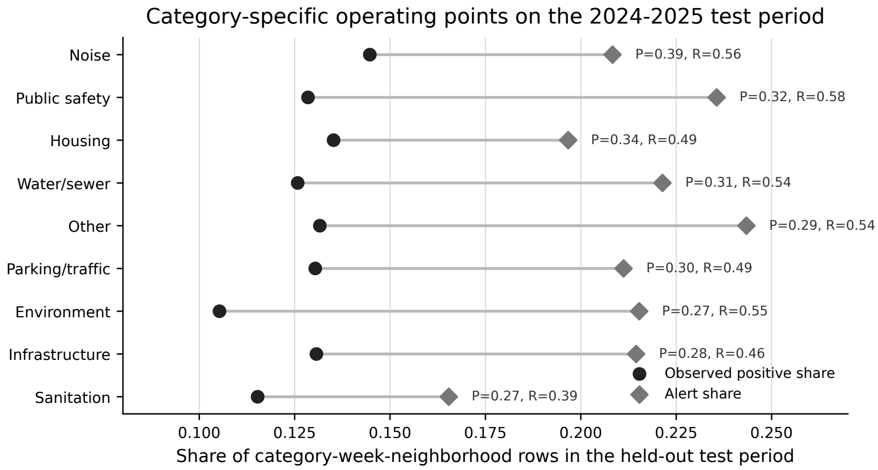

Fig. 2 Observed positive share and validation-threshold alert share by complaint category in the held-out test period. Labels show category precision and recall at the selected operating point.

and the top 10% has precision 0.3866. These concentration results position the model as prioritization support rather than an automatic allocation rule.

4.2 Category-Level Alert Analysis

Performance varies across municipal service domains (Table 5). Noise has the strongest F1, followed by public safety, housing, and water/sewer. Sanitation, infrastructure, and environment are weaker by F1, although several retain moderate recall. Positives in Table 5 are observed abnormal next-week rows within the held-out test set; alerts are rows whose ensemble score exceeds the validation-selected threshold for that complaint category. Alert counts therefore need not be proportional to positives. They depend on the category threshold, the category score distribution, separation between positive and negative rows, and the validation operating point. The counts are consistent with the test confusion matrix: summing the category positives gives 31,570 and summing the category alerts gives 52,594.

Table 6 Sensitivity to abnormal-demand definition under a compact LightGBM protocol. Scores are for within-table comparison and are not directly comparable with the final full-training ensemble in Table 4.

Target definition	Pos. share	F1	PR-AUC	ROC-AUC
4w + 1.5σ	0.1443	0.4156	0.3659	0.7716
8w + 1.0σ	0.1833	0.4411	0.3982	0.7505
8w + 1.5σ (reference)	0.1275	0.3753	0.3214	0.7608
8w + 2.0σ	0.0885	0.3178	0.2568	0.7706
12w + 1.5σ	0.1180	0.3665	0.3155	0.7677

Figure 2 makes the operating-point trade-off visible. Noise has more observed positives than public safety but fewer alerts; its higher threshold yields higher precision and slightly lower recall, while public safety issues more alerts to capture higher recall at lower precision. This is not a data contradiction, but a consequence of category thresholds and score distributions. Relative to a global ensemble threshold, the category-specific ensemble has macro-F1 0.3821 versus 0.3838, macro precision 0.3069 versus 0.2989, macro recall 0.5107 versus 0.5394, and mean alert rate 0.2124 versus 0.2305. The result should be read as the pre-specified category-specific strategy selected on validation, not as the best aggregate-F1 strategy.

Target-definition sensitivity was also checked with a compact LightGBM protocol using the full context features, the same chronological split, and validation-selected thresholds (Table 6). Event prevalence changes as expected: a 1.0-standard-deviation threshold marks more rows positive, while 2.0 standard deviations yields a rarer target. The reference 8-week, 1.5-standard-deviation target definition sits between these alternatives. F1 and PR-AUC move with prevalence, but ROC-AUC remains in a narrow 0.7505–0.7716 range across the tested definitions, suggesting that the ranking problem is not an artifact of a single window/multiplier choice.

4.3 Explanation Results

Figure 3 summarizes SHAP importance by feature group for the tuned LightGBM component. Historical temporal features dominate, accounting for 58.96% of summed mean absolute SHAP importance across 21 features. Service category accounts for 10.73% across nine indicators, PLUTO built environment for 10.18% across 53 features, OSM POI context for 8.13% across 103 features, weather for 4.88% across 22 features, calendar features for 4.77% across four features, and current demand for 1.61% for one feature. Because these are summed group importances, groups with more features can accumulate larger total contribution.

Figure 4 shows the leading individual features. The strongest terms are historical: `rolling_8w_std`, `ratio_to_8w_mean`, `lag_52w_count`, `rolling_4w_mean`, and `lag_1w_count`. Calendar timing appears through `week_of_year`, current demand appears through `complaint_count`, and several complaint-category indicators enter the top ranks. These patterns align with the ablation results: recent history and seasonality carry most of the signal, while context and category terms refine the score. SHAP explains the LightGBM component, not every ensemble alert, and should be read as predictive association rather than causal effect.

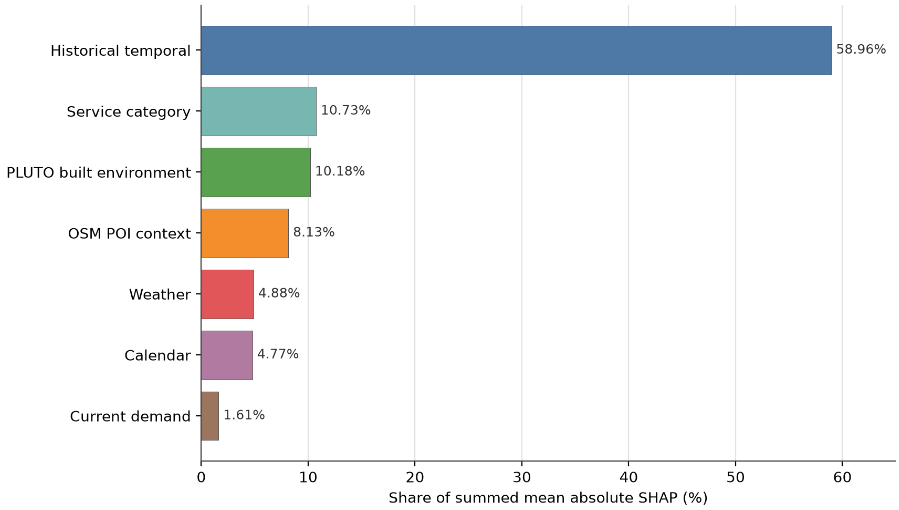

Fig. 3 Normalized summed mean absolute SHAP importance for the tuned LightGBM explanation backbone. Historical temporal demand features dominate, while service-category, PLUTO, OSM POI, weather, and calendar features add smaller but interpretable contextual contributions.

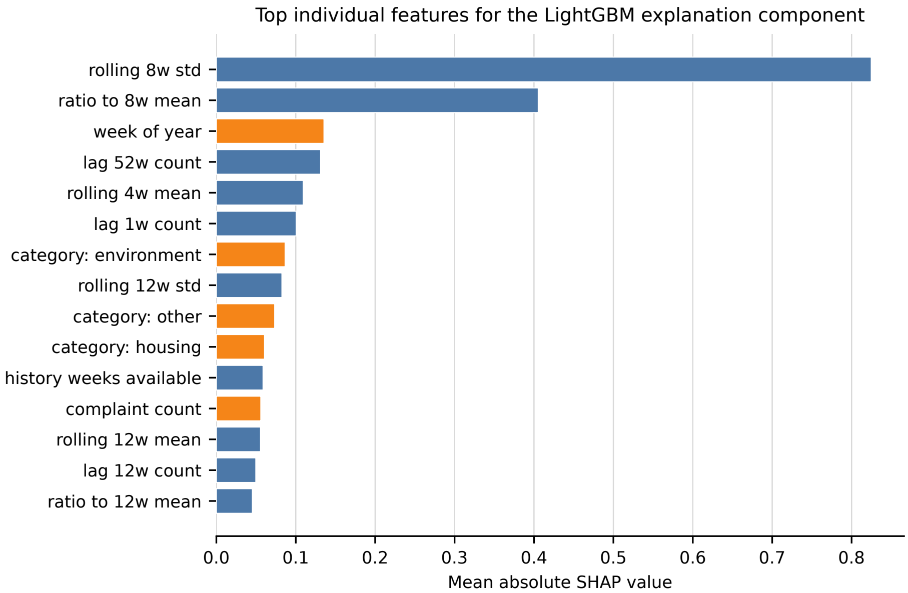

Fig. 4 Top individual SHAP features for the tuned LightGBM explanation component. Blue bars are historical temporal features; orange bars are other feature groups.

Two local examples illustrate use: a true-positive housing alert is associated with eight-week volatility, seasonal timing, and residential built-environment features, while a false-positive public-safety alert is lifted by annual and recent lag counts. These cases support analyst review but do not establish causal effects.

5 Discussion

5.1 Operational and Methodological Implications

The model is most useful as early-warning decision support. Recall of 0.5117 means it captures about half of future abnormal reported-demand events, while precision of 0.3071 means many alerts are false positives. Agencies should therefore use alerts to prioritize review, pre-position attention, or monitor risk queues, with final actions incorporating severity, staffing, and local knowledge.

Category thresholds make the alert layer more transparent because service domains do not share the same base rates, volatility, response costs, or tolerance for missed events. A sanitation alert and a public-safety alert can have different consequences even when their model scores are similar. The small aggregate-F1 difference between global and category-specific thresholds reinforces the main point: the benefit is explicit service-domain trade-off control, not a claim of universal metric dominance. The ablation results also qualify the role of urban context. Calendar features provide the largest gain over the core configuration, while OSM and PLUTO add complementary ranking signal. Sensitivity checks show that target prevalence changes under alternative abnormal-demand definitions, so thresholds and workload should be revisited if an agency adopts a different event definition.

5.2 Limitations and Future Work

Several limits should guide interpretation. NYC 311 data measure reported service demand, not all underlying urban problems, and reporting propensity can reproduce socio-spatial inequities. OSM POI and PLUTO are static 2026 snapshots used across a 2015-2025 panel, so they carry snapshot/look-ahead risk and should be interpreted only as broad context. NOAA weather is weekly city-level exposure rather than NTA microclimate. Hyperparameters are selected from sampled tuning runs, although final reported scores are rebuilt on all training rows; three sampled seeds produced test F1 0.3817 ± 0.0021, PR-AUC 0.3313 ± 0.0025, and ROC-AUC 0.7678 ± 0.0007. Category thresholds are optimized for validation F1, not agency-specific costs, and full operational utility would require response-time, staffing, service-outcome, and fairness data. Future work should test historical or time-matched OSM and PLUTO, spatial weather, calibrated probabilities, cost-sensitive thresholds, alternative target definitions, cross-city transfer, and operational outcome evaluation.

6 Conclusion

An explainable forecasting framework was developed for next-week abnormal reported demand in NYC 311 data at the neighborhood, week, and service-category level. The

NTA-week-category formulation, multi-source context, and target-week chronological split frame 311 forecasting as an early-warning task rather than generic count prediction. Results from the held-out future period show that the LightGBM/XGBoost ensemble captures about half of abnormal reported-demand events, while ablation and SHAP evidence indicate that historical temporal dynamics dominate and contextual features add smaller complementary signal. Category thresholds make service-domain alert trade-offs explicit. The main limitation remains central: 311 data forecast reported demand, not the complete distribution of urban need.

Declarations

Data and code availability. Source data are NYC Open Data 311 historical archive `76ig-c548`, NYC Open Data 311 current archive `erm2-nwe9`, NOAA GHCN-Daily station `GHCND:USW00094728`, OpenStreetMap Overpass, NYC PLUTO/MapPLUTO `64uk-42ks`, and 2020 NTA boundaries `9nt8-h7nd`; the local snapshot was extracted on 2026-06-25. Scripts, mappings, configurations, and evaluation code are available at `https://github.com/PhongLam26/SIMC_NYC`; processed data release is subject to licensing, redistribution, and storage constraints.

Competing interests. The authors declare no competing interests.

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Author contributions. Tran Dai Phong Lam led data processing, modeling, analysis, and drafting; Thu Le served as academic advisor and supervised the study design, methodology, and manuscript editing; Nguyen Quoc Hung and Nguyen Trung Trinh contributed validation and review.

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